

DCP-DualMask: A Physically Guided Continual Test-Time Adaptation Framework for Image Dehazing in Dynamic Scenes

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Abstract—Image dehazing is a critical task for numerous high-level vision applications, yet deep learning-based methods often struggle when deployed in dynamic real-world environments due to a persistent domain shift between synthetic training data and authentic test scenes. Existing test-time adaptation methods, such as entropy minimization, fail to adequately address the spatial heterogeneity of haze, frequently leading to error accumulation in dense haze regions. To overcome this limitation, we propose DCP-DualMask, a novel continual test-time dehazing framework. Our approach introduces a physically guided dual-mask strategy inspired by masked image modeling. Rather than using random masking, we leverage the dark channel prior to localise regions of high haze uncertainty and combine this with superpixel segmentation to preserve local structures. This forms a challenging self-supervised reconstruction task that encourages the model to infer occluded scene content from contextual cues. A dual-mask mechanism further applies a soft mask in the input space to simulate visibility loss and a hard mask in the feature space to enhance semantic robustness. A consistency loss enforces local prediction smoothness. Experimental results demonstrate that our method effectively mitigates domain shift, achieving state-of-the-art average PSNR/SSIM of 14.56 dB/0.64 across five real-world haze datasets, significantly outperforming existing approaches in both dehazing performance and generalisation under dynamic open scenarios. This work provides a robust and generalisable solution for continual test-time adaptation in image dehazing, paving the way for more reliable vision systems operating in uncontrolled environments.

Index Terms—image dehazing, test-time adaptation, masked image modeling, dark channel prior, domain shift

I. INTRODUCTION

Image dehazing is a fundamental low-level vision task for applications such as autonomous driving, surveillance, and remote sensing. Although deep learning-based methods perform well on synthetic datasets, their performance often degrades in real-world open scenarios due to continuously varying haze and imaging conditions, resulting in a significant synthetic-to-real domain gap. In this paper, “dynamic scenes” refer to such real-world test environments with varying haze and imaging conditions.

Test-time adaptation can mitigate this gap using unlabeled test data, but existing CTTA methods are not directly suitable for dehazing because dehazing requires pixel-level texture

restoration and real haze is spatially non-uniform. Dense-haze regions may produce unreliable predictions and cause error accumulation in pseudo-label-based adaptation.

Inspired by masked image modeling (MIM) [1], we propose DCP-DualMask, a physically guided CTTA framework that uses the dark channel prior (DCP) and SLIC superpixels to construct haze-aware masks for dense-haze regions. In the teacher–student framework, the student receives the complete hazy image, while the teacher uses masked inputs to provide stable pseudo-supervision. The proposed input-level soft mask and feature-level hard mask jointly improve robustness under real-world domain shifts. Finally, we introduce a superpixel consistency loss to further suppress artifacts in dehazing results by exploiting local smoothness constraints within superpixels. The main contributions of this paper are as follows:

- 1) A physically guided continual test-time adaptation framework for image dehazing in dynamic scenes is proposed. By introducing structured-mask consistency learning guided by the dark channel prior, the framework improves output stability and robustness under continual domain shift.
- 2) A superpixel masking strategy guided by the dark channel prior is designed. This strategy uses physical priors to localize dense-haze regions and preserves local structures through superpixel segmentation, forcing the model to learn robust context-dependent dehazing ability.
- 3) A dual-mask consistency learning mechanism is proposed. By combining multi-scale perturbations at the input level and the feature level, it significantly enhances the generalization ability and robustness of the dehazing model in open scenarios.

II. RELATED WORK

Existing dehazing methods can be broadly divided into physical-prior-based and deep-learning-based approaches. Physical-prior-based methods use statistical assumptions from natural images to constrain the atmospheric scattering model. A representative example is the dark channel prior proposed by He et al. [2], which estimates transmission based on low-intensity pixels in haze-free non-sky regions. The color attenu-

ation prior further models the relationship between brightness and saturation. Although effective under ideal assumptions, these priors may fail in atypical scenes such as sky regions or white objects, causing biased transmission estimation and artifacts. Nevertheless, their physical constraints remain useful for guiding deep networks on unlabeled real hazy images.

With convolutional neural networks, image dehazing has shifted toward data-driven end-to-end learning. Methods such as FFA-Net [3] achieved strong results on synthetic datasets, and Transformer-based models further improved detail restoration through long-range modeling. However, these methods are typically trained on synthetic data and often degrade in real-world haze due to the synthetic-to-real domain gap. Although recent works improve generalization by introducing priors or redesigning network architectures, most still rely on offline retraining or extra assumptions, limiting dynamic adaptation during inference.

Continual test-time adaptation (CTTA) updates models online using unlabeled test data and is suitable for dynamic real-world environments. CoTTA [4] uses a mean-teacher framework and random restoration to address distribution drift and reduce catastrophic forgetting. However, pseudo-label-based self-training can produce unreliable supervision under severe domain shifts. Recent TTA methods introduce masked image modeling to infer missing content through occlusion-reconstruction, providing a context-consistency constraint that is less dependent on prediction confidence and more robust to dynamic shifts.

III. METHOD

A. Overall Architecture

The proposed physically guided dual-mask continual test-time adaptive dehazing framework is shown in Fig. 1. It mainly consists of a dual-mask generation module and a teacher-student self-training module. A base dehazing network pretrained on synthetic data is used to initialize both the student and teacher networks. During testing, the dual-mask generation module constructs masked inputs from each unlabeled real hazy image by combining the dark channel prior with superpixel structural information. The student network receives the original hazy image, while the teacher branch takes the masked input and the corresponding feature mask to produce stable pseudo-supervision. The student is optimized by a consistency loss between student and teacher outputs, and the teacher network is updated using an exponential moving average (EMA) strategy to maintain adaptation stability.

B. Dark-Channel-Guided Superpixel Mask Selection Strategy

Real-world haze is spatially non-uniform, and dense-haze regions often contain severe information loss and pseudo-textures. To avoid structural damage caused by conventional masks, we combine SLIC superpixels with the dark channel prior to select high-haze regions in a structure-preserving manner.

1) *Pixel-level Prior Extraction*: First, in order to quantify haze concentration distribution at the physical level, we compute the dark channel map $D(x)$ for the input image I_{haze} . According to the theory of the dark channel prior, the pixel-level dark channel map is calculated as follows:

$$D(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{R, G, B\}} I_{\text{haze}}^c(y) \right) \quad (1)$$

where $D(x)$ denotes the dark-channel value at pixel x ; $I_{\text{haze}}^c(y)$ denotes the intensity of the input image at hazy position y in color channel c selected from the red, green, and blue channels; and $\Omega(x)$ denotes a local window centered at pixel x . Although $D(x)$ provides pixel-level haze-concentration information, because it is computed from a local window, it lacks perception of the overall object structure.

2) *Structure Segmentation*: To introduce structural constraints, we independently perform superpixel segmentation on the original image I_{haze} using the SLIC (Simple Linear Iterative Clustering) algorithm. SLIC partitions the image into K semantically coherent regions according to color and spatial distances, producing the label map S :

$$S(x) = \text{SLIC}(I_{\text{haze}}) \quad (2)$$

where $S(x)$ denotes the generated superpixel label with the same size as the original image. For any pixel x in the image, its corresponding label satisfies $S(x) \in \{0, 1, \dots, K-1\}$, which indicates the superpixel index to which the pixel belongs. K is the preset total number of superpixels. This step uses the edge information of the original image to ensure that the texture and structure inside each region are consistent.

3) *Region-based Guidance Fusion and Patch Map Generation*: After obtaining the pixel-level clue $D(x)$ and the region-level structure $S(x)$, we fuse them by region mean pooling. For the k -th superpixel region R_k , the mean value of the dark channel inside the region is computed as the ‘‘haze concentration score’’ μ_k of that region:

$$\mu_k = \frac{1}{|R_k|} \sum_{x \in R_k} D(x) \quad (3)$$

where $R_k = \{x \mid S(x) = k\}$ denotes the set of all pixels whose label is k ; $|R_k|$ denotes the total number of pixels in R_k , that is, the area of the superpixel; and μ_k is the aggregated haze-concentration confidence of that region. Subsequently, the region score μ_k is filled back to the corresponding pixel positions to generate the structured haze density map, namely $H(x) = \mu_{S(x)}$. At this point, $H(x)$ not only preserves the sensitivity of physical priors to haze concentration, but also possesses the edge-preserving property of superpixels.

4) *Top-ratio Superpixel Masking*: Finally, in order to screen out high-haze regions that need to be emphasized at different levels from the region-level haze density map, this paper adopts a Top-ratio selection strategy. We treat the haze-concentration scores of the superpixel regions, $\{\mu_k\}$, as a quantitative indicator of degradation degree, sort all region scores in descending order, and select the top P superpixel

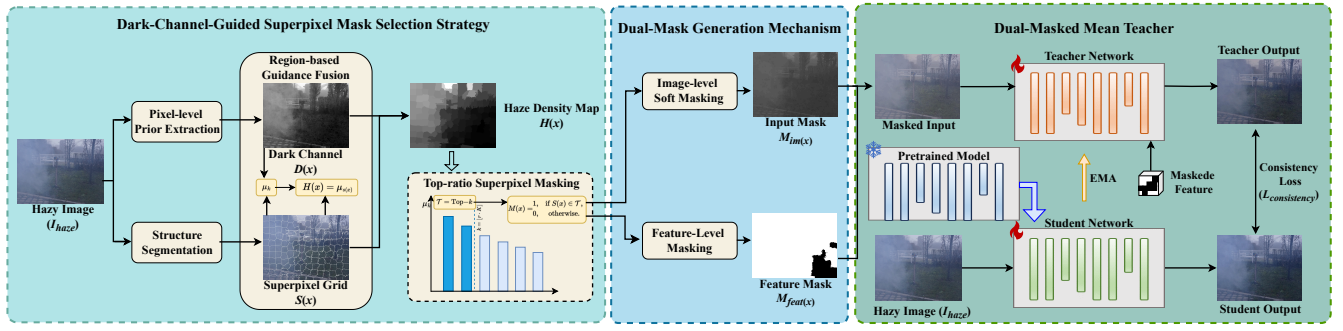


Fig. 1. Image dehazing framework based on dark-channel-prior-guided dual masking.

set $\mathcal{T}$ corresponding to ratio r to generate the binary mask $M(x)$:

$$M(x) = \begin{cases} 1, & \text{if } S(x) \in \mathcal{T}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

where the set satisfies $\mathcal{T} = \text{Top-}k(\{\mu_k\})$, with $k = \max(1, \lfloor r \cdot K \rfloor)$. Through this strategy, a certain proportion of high-haze regions can be flexibly selected from all K superpixels according to the preset ratio r .

Eq. (4) provides a unified definition for the proposed dual mask. Specifically, when $r = r_{im}$, $k = k_{im}$, and $\mathcal{T} = \mathcal{T}_{im}$, it generates the input-level mask $M_{im}(x)$; when $r = r_{feat}$, $k = k_{feat}$, and $\mathcal{T} = \mathcal{T}_{feat}$, it generates the feature-level mask $M_{feat}(x)$, where $k_{im} = \max(1, \lfloor r_{im} \cdot K \rfloor)$ and $k_{feat} = \max(1, \lfloor r_{feat} \cdot K \rfloor)$. Thus, the two masks are generated from the same dark-channel-guided superpixel haze-score ranking but use different selection ratios to serve input-level and feature-level perturbations, respectively.

In our implementation, r_{im} and r_{feat} are fixed hyperparameters rather than image-adaptive or learnable parameters. However, the spatial locations of the masked regions remain image-adaptive because the selected superpixels are determined by the haze-score ranking of each individual test image.

C. Dual-Mask Generation Mechanism

After determining “where to occlude” through the dark-channel-guided superpixel selection strategy, we further design the dual-mask generation mechanism to specify “how to occlude.” Specifically, we compute the dark channel map, perform SLIC superpixel segmentation, calculate the average dark-channel value of each superpixel as its haze score, and sort all superpixels by their haze scores. According to two mask ratios, r_{im} and r_{feat} , the top-ranked high-haze superpixels are selected to generate the input-level mask M_{im} and the feature-level mask M_{feat} , respectively. The input-level mask is implemented as a soft mask to simulate visibility attenuation while preserving local structures, whereas the feature-level mask is implemented as a hard mask to suppress unreliable feature activations in dense-haze regions and enhance teacher-student consistency learning.

1) *Image-level Soft Masking*: To avoid abrupt boundaries and color discontinuities caused by conventional hard occlusion (directly setting values to zero), this paper constructs the soft-mask weight $w(x)$ within the support domain of M_{im} , so that the selected regions are assigned different occlusion intensities from weak to strong according to haze concentration. The weight $w(x)$ is defined as follows:

$$w(x) = \begin{cases} \hat{w}(\text{clip}(H(x))), & M_{im}(x) = 1, \\ 0, & M_{im}(x) = 0. \end{cases} \quad (5)$$

where $\hat{w}(\cdot)$ denotes the soft-mask weight assigned to the selected region. Specifically, for pixels satisfying $M_{im}(x) = 1$, the mask weight is computed by applying a clipped nonlinear mapping to the normalized haze-concentration value $H(x)$. The mapping starts from the lower bound $m_{\min}$ and increases with haze concentration, with the increment range controlled by $m_{\max} - m_{\min}$ and the curve shape controlled by γ . The resulting value is finally truncated to the interval $[0, m_{\max}]$. $\text{clip}(\cdot)$ denotes the truncation function.

Based on this weight, the teacher branch performs input-level suppression on the enhanced input I_{aug} to obtain the teacher input $I_{teacher}$:

$$I_{teacher} = I_{aug} \odot (1 - w(x)) \quad (6)$$

This soft-mask design simulates the gradual loss of local visibility caused by medium scattering in real scenes and forces the network to focus on global contextual structures.

2) *Feature-Level Masking*: To further suppress the propagation of pseudo-textures from dense-haze regions in deep networks, this paper applies a feature-level mask to the shallow features of the teacher network. Specifically, the feature-level hard mask is applied only to the EMA teacher branch. In implementation, we register a forward hook on the first Conv2d layer of the teacher network and apply the hard mask to the output feature map of this layer. The student branch receives the original unmasked hazy image and does not use the feature-level hard mask. Let the output feature of a certain shallow layer of the teacher network be $F \in \mathbb{R}^{B \times C \times H_f \times W_f}$. We downsample the feature-level binary mask M_{feat} to the feature scale by nearest-neighbor interpolation to obtain M_f .

Different from the soft processing at the input level, a channel-mean replacement strategy is adopted here. We compute the channel mean of the masked region as follows:

$$\bar{F} = \frac{1}{H_f W_f} \sum_{i=1}^{H_f} \sum_{j=1}^{W_f} F(:, :, i, j) \quad (7)$$

and then generate the processed feature F' :

$$F' = F \odot (1 - M_f) + \bar{F} \odot M_f \quad (8)$$

where F denotes the feature map extracted by the encoder, M_f denotes the feature-level mask, and $\bar{F}$ denotes the spatial average of F . Here, $\bar{F}$ is spatially broadcast to the feature size before replacement. By replacing unreliable high-frequency details with smooth global statistics, this design makes the teacher network output more stable pseudo-supervision signals in dense-haze regions, thereby guiding the student network to learn a dehazing mapping that is more robust to dynamic haze changes.

D. Loss Function

Without label supervision during testing, a consistency loss is adopted to constrain the student prediction toward the teacher output, improving adaptation robustness and generalization. The consistency loss is defined as follows:

$$\mathcal{L}_{\text{consistency}} = \|P_t - P_s\|_1 \quad (9)$$

where P_t denotes the prediction result of the teacher network, P_s denotes the prediction result of the student network, and $\|\cdot\|_1$ denotes the L1 loss.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Evaluation Metrics

To evaluate the effectiveness and generalization ability of the proposed method in real hazy scenes, we conduct experiments on five public datasets, including NH-HAZE [5], Dense-Haze [6], O-HAZE [7], I-HAZE [8], and the indoor haze subset of Weather Bench [9]. These datasets cover indoor and outdoor scenes, homogeneous and non-homogeneous haze, as well as thin and dense haze. Since this study focuses on image dehazing, only the indoor haze subset is selected from Weather Bench. We use PSNR [10] and SSIM [11] as evaluation metrics. PSNR measures pixel-level reconstruction fidelity, while SSIM evaluates structural similarity between the dehazed image and the reference image.

B. Experimental Comparison

To verify the effectiveness and generalization ability of the proposed method on real hazy scenes, comparative experiments are conducted against seven SOTA methods, including five dehazing methods, FFANet [3], C2PNet [12], Dehamer [13], DEANet [14], and WLDNet [15], as well as two typical test-time adaptation methods, CoTTA [4] and ViDA [16].

TABLE I
QUANTITATIVE COMPARISON ON REAL-WORLD HAZE DATASETS.

Method	NH P/S	Dense P/S	O P/S	I P/S	WB P/S	Avg. P/S
FFANet	11.94/0.47	10.36/0.44	15.70/0.66	16.21/0.77	15.52/0.75	13.95/0.62
Dehamer	12.04/0.46	11.00/0.43	15.61/0.67	16.57/0.79	16.09/0.74	14.26/0.62
C2PNet	12.27/0.49	11.05/0.45	15.95/0.67	16.57/0.79	15.44/0.75	14.26/0.63
DEANet	12.29/0.50	11.40/0.46	16.50/0.68	16.47/0.79	15.29/0.76	14.39/0.64
WLDNet	12.05/0.52	11.25/0.38	16.09/0.74	16.49/0.76	14.96/0.65	14.17/0.61
CoTTA	12.03/0.47	11.18/0.45	16.00/0.65	16.48/0.79	16.22/0.77	14.38/0.63
ViDA	12.06/0.48	10.96/0.45	15.89/0.69	16.63/0.79	15.82/0.77	14.27/0.63
Ours	12.26/0.48	11.43/0.46	16.20/0.69	16.71/0.80	16.18/0.77	14.56/0.64

P/S denotes PSNR/SSIM. NH, Dense, O, I, and WB denote NH-HAZE, Dense-Haze, O-Haze, I-Haze, and Weather Bench (indoor), respectively.

TABLE II
ABLATION STUDY OF DUAL-MASK AND SUPERPIXEL STRATEGIES.

Method	NH P/S	Dense P/S	O P/S	I P/S	WB P/S	Avg. P/S
basic	12.06/0.47	10.99/0.45	15.83/0.68	16.68/0.80	16.09/0.77	14.33/0.64
+image Mask	12.14/0.48	11.37/0.46	15.90/0.68	16.62/0.80	16.01/0.77	14.41/0.64
+feature Mask	12.12/0.48	11.33/0.46	16.05/0.69	16.65/0.80	16.10/0.77	14.45/0.64
+dual Mask	12.15/0.48	11.32/0.46	16.08/0.69	16.62/0.79	16.15/0.77	14.46/0.64
+dual Mask+SP	12.26/0.48	11.43/0.46	16.20/0.69	16.71/0.80	16.18/0.77	14.56/0.64

P/S denotes PSNR/SSIM. NH, Dense, O, I, and WB denote NH-HAZE, Dense-Haze, O-Haze, I-Haze, and Weather Bench (indoor), respectively.

1) *Quantitative Comparison:* Table I reports the PSNR/SSIM comparison results on five real hazy-image datasets. The proposed method achieves the best average PSNR of 14.56 dB and a competitive average SSIM of 0.64, indicating its ability to improve pixel-level fidelity while preserving structural consistency. Compared with supervised dehazing networks, our method improves the average PSNR by 0.61 dB, 0.30 dB, 0.30 dB, 0.17 dB, and 0.39 dB over FFANet [3], Dehamer [13], C2PNet [12], DEANet [14], and WLDNet [15], respectively. It also outperforms the test-time adaptation methods CoTTA [4] and ViDA [16] by 0.18 dB and 0.29 dB in average PSNR, demonstrating better cross-dataset robustness under real-world haze variations.

2) *Qualitative Comparison:* Figure 2 presents visual comparisons on two real hazy scenes, with red boxes showing enlarged regions. Under heavy haze, the input images suffer from low contrast, blurred contours, and compressed texture details. Although supervised dehazing and test-time adaptation methods improve visibility to some extent, they still exhibit limited detail restoration or unstable local contrast. In contrast, the proposed method produces more stable visual results, better restores distant structures and road textures, and maintains more natural brightness, color distribution, and regional consistency.

C. Ablation Experiments

To evaluate the contribution of each component, we conduct ablation experiments on five real hazy-image datasets. The baseline, denoted as basic, is Dehamer combined with the



Fig. 2. Visual comparison of different dehazing and test-time adaptation methods on real-world hazy scenes.

TABLE III
IMPACT OF DUAL-MASK PERTURBATION INTENSITIES.

r_{im}	r_{feat}	NH P/S	Dense P/S	O P/S	I P/S	WB P/S	Avg. P/S
0.3	0.7	12.33/0.47	11.43/0.46	15.98/0.68	16.69/0.80	16.01/0.77	14.49/0.64
0.7	0.3	12.29/0.46	11.49/0.45	15.86/0.67	16.60/0.79	16.28/0.77	14.50/0.63
0.5	0.5	12.30/0.47	11.48/0.46	16.05/0.68	16.68/0.80	16.02/0.77	14.51/0.64
0.9	0.1	12.33/0.46	11.53/0.45	15.77/0.66	16.58/0.79	15.94/0.77	14.43/0.63
0.1	0.9	12.26/0.48	11.43/0.46	16.20/0.69	16.71/0.80	16.18/0.77	14.56/0.64

P/S denotes PSNR/SSIM. NH, Dense, O, I, and WB denote NH-HAZE, Dense-Haze, O-Haze, I-Haze, and Weather Bench (indoor), respectively.

Mean-Teacher framework. Image Mask and feature Mask introduce masking constraints in the input and feature spaces, respectively; Dual Mask combines both, and Dual Mask+superpixel further adds superpixel refinement to enhance regional consistency.

As shown in Table II, the average PSNR increases from 14.33 dB for the baseline to 14.41 dB, 14.45 dB, and 14.46 dB after introducing Image Mask, feature Mask, and Dual Mask, respectively. With superpixel refinement, the full model achieves the best performance of 14.56/0.64, improving PSNR by 0.23 dB over the baseline. These results demonstrate the effectiveness of the proposed dual-mask and superpixel strategies.

D. Parameter Sensitivity Analysis

To analyze the influence of perturbation intensity, we evaluate different combinations of the image-level mask ratio r_{im} and the feature-level mask ratio r_{feat} on five real hazy-image datasets. As shown in Table III, different ratio settings lead to moderate performance variations, indicating that the balance between input-level and feature-level perturbations affects dehazing quality. Overall, the average PSNR remains within 14.43–14.56 dB and SSIM remains within 0.63–0.64, showing that the proposed method is relatively robust to hyperparameter changes. Among all settings, the 0.1/0.9 combination achieves the best average performance of 14.56/0.64.

V. CONCLUSION

This paper proposes DCP-DualMask, a physically guided continual test-time adaptation framework for dynamic image dehazing. By combining dark-channel-prior-guided dual masking with continual adaptation, the framework mitigates the synthetic-to-real domain gap and improves dehazing quality, detail restoration, and robustness on real-world haze benchmarks.

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